UG (NEP) 4th SEMESTER
COMPUTER APPLICATIONS – DBMS
Course Code: CAP422J1

Session: June 2025 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 75 Minimum Marks: 27

NOTE: Attempt all questions from Sections A and B and only two questions from Section C.

SECTION A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

1. Define entity and entity set?
2. Name any two DDL commands?
3. What is a data model?
4. What is a foreign key?
5. Name any two aggregate functions in SQL?
6. Define locking?
7. What is data abstraction?
8. What is a transaction?

SECTION B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. What are the different types of data models?

OR

Explain the concept of schema and instances.

3. What is a relational data model? Explain with an example.

OR

Draw an ER diagram for a college library management system.

4. Describe 3NF with a suitable example.

OR

Explain functional dependencies and their use in normalization.

5. Explain aggregate functions in SQL with example.

OR

Explain SELECT command with WHERE and LIKE clause.

SECTION C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Discuss three-schema architecture and data independence in detail.
7. Given the unnormalized relation: Student(RollNo, SName, Courses(CourseID, CName), AdvisorID, AdvisorName):
   a) Convert the relation into 1NF. b) Decompose it further to 2NF. c) Justify each step of normalization.
8. Describe DDL, DML, and DCL with SQL examples.
9. Explain concurrency control techniques, focusing on 2-phase locking and timestamp ordering.